PPchart Usage Notes

PPchart is a Python program to create PDF files relating to Planetary Phenomena. It is invoked using a name that runs Python (e.g. py, python, python3), the program name (PPchart.py) and optional arguments, some of which require a value. A typical command line could look like:

`py  PPchart.py -tex -pv2 -dpo -lat 45.0N`

('**`-lat`**' takes the value '**`45.0N`**' above)

Entering an invalid argument prints out a brief list of all permitted arguments:

```
Valid command line arguments are:
 -v   ... 'verbose': to send detailed output to the terminal
 -vt  ... 'verbose': to send pdfTeX output to the terminal
 -log ... to keep the log file
 -tex ... to keep the tex file
 -a4  ... A4 papersize
 -let ... Letter papersize
 -dpo ... data pages only
 -un  ... also plot Uranus & Neptune*
 -pss ... plot sunrise/sunset & civil dusk/dawn at chosen latitude*
 -sbr ... use square brackets in Unix filenames
 -sp  ... execute in single-processing mode (slower)
      * applies only to the Planet Diagram
 --------  Planet Visibility charts  --------
 -pv1 ..... for Mercury
 -pv2 ..... for Venus
 -pv3 ..... for Mars
 -pv4 ..... for Jupiter
 -pv5 ..... for Saturn
 -pv6 ..... for Uranus
 -pv7 ..... for Neptune
 -pv  ..... for the first 5 planets (Mercury to Saturn)
 -lat 51.5N ... select one latitude as nn.nN or nn.nS
 -lat ... print charts with all the following latitudes:
          60S 58S 55S 50S 45S 40S 30S 15S 0 15N 30N 40N 45N 50N
          55N 58N 60N 62N 63N 64N 65N 66N 67N 68N 69N 70N 71N 72N
 -lats .. print all above latitudes as separate PDF chart files
 -orth .. orthogonal plot data: data 00h to 24h is aligned vertically
          on start of day
```

Latitudes are confined to the range found in Nautical Almanac data: 60°S to 72°N.

Some arguments have no meaning in the chosen context, e.g. **-pss** and **-un** have no meaning in Planet Visibility charts. The sequence of arguments is irrelevant; however, an argument's value must follow the argument, e.g. '**-lat 45.0N**' cannot be separated.

The year or range of years are then requested as follows:

```
Enter as numeric digits:
   - 'YYYY'        (1951 <= year <= 2050)
   - 'YYYY-YYYY'   for a range of years
   - nothing for the current year
```

PPchart can be used as follows:

- To create Planetary Phenomena pages at the default latitude 51.5°N:
  py PPchart.py
- To create Planetary Phenomena pages at latitude 54.1°N:
  py PPchart.py -lat 54.1N
- Ditto, including sunrise/sunset and civil dusk/dawn offset from Sun's Meridian Passage at the chosen latitude[1]:
  py PPchart.py -lat 54.1N -pss
- Ditto, without the initial description text:
  py PPchart.py -lat 54.1N -pss -dpo
- To create Planet Visibility Charts for Venus at latitude 41.5°N:
  py PPchart.py -pv2 -lat 41.5N
- To create Planet Visibility Charts for Mars at all latitudes in one PDF:
  py PPchart.py -pv3 -lat
- To create Planet Visibility Charts for Mars at all latitudes in separate PDF files:
  py PPchart.py -pv3 -lats
- Ditto, without the initial description text:
  py PPchart.py -pv3 -lats -dpo

[1] This is slightly confusing to grasp as one assumes that the Sun's Meridian Passage now becomes Sunrise and the red line below represents Civil Dawn as minutes (time) offset from, i.e. before, Sunrise. Alternatively, one assumes that the Sun's Meridian Passage now becomes Sunset and the blue line above represents Civil Dusk as minutes (time) offset from, i.e. after, Sunset. Thus, as a *preliminary assumption*, planets between the red and blue lines are not visible on those days. This is the basic rationale for the shaded area 45 minutes (time) above and below the Sun's Meridian Passage on every Planet Diagram, though this does not take the latitude into account.

Extended functionality for developers*

Master Reference data for Planet Visibility charts can be created for use with Regression Testing – the advantage being that the created .TEX files can be compared without creating any PDF charts. This saves time in Regression Testing; however identical data must be generated. The Master Data includes PDF files for reference. The argument '-tex' is applied by default – and so is '-dpo'.

The Master Data contains only orthogonal charts as these are preferential for debugging discrepancies. (Orthogonal charts employ time axes at 90° to each other, which are technically incorrect as the chart upper border at 24h has already advanced by 1 day to 00h the next day.)

The folder structure created is as follows, i.e. the Master Data structure is <u>two folder levels</u> *above* the folder in which PPchart executes:

- Top level folder
    - _2000
    - _2001
    - _2002
    - _2003
    - _2004
    - code_execution folder
        - source files in which to execute PPchart code
            - _rt (folder with regression test results)

Missing folders are created automatically. Sample commands to create (or update) Master Data (the first four examples create one PDF file per latitude):

- Create Mercury 67°N in the year 2000:
  `py PPchart.py -pv1 -lat 67.0N -nm 00`
- Create Venus at all latitudes in the year 2001:
  `py PPchart.py -pv2 -lats -nm 01`
- Create all planets at all latitudes in the year 2003:
  `py PPchart.py -pv -lats -nm 03`
- Create first five planets at all latitudes in the years 2003 to 2004:
  `py PPchart.py -pv -lats -nm 03-04`
- Create all latitudes for the first five planets in the years 2003 to 2004 <u>in one file per year</u>:
  `py PPchart.py -pv -lat -nm 03-04`

The last example above (with all latitudes in one PDF file) is not used for Regression Testing: the PDF file is a quick way to visually verify correct chart creation for all latitudes.

Note: The **`-nsb`** argument includes **`-nsdbh`**.

* Extended functionality is enabled by setting **devuser = True** in the main program.

Planet Visibility Regression testing can be performed with PPchart. This assumes that correct .tex files have been saved in a folder structure named '_year' two levels higher. Regression testing is performed by creating only the .tex files and comparing them to a reference master without the need to create a PDF chart, which saves time. It requires identical astronomical data, so set 'useIERS = False' in *config.py* and ensure the same version of Skyfield is used (i.e. with the same version of the built-in UT1 tables). The .tex files are not deleted – they are there for debugging.

- Test Mercury 67°N in the year 2000:
  `py PPchart.py -pv1 -lat 67.0N -rt 00`
- Test Venus at all latitudes in the year 2001:
  `py PPchart.py -pv2 -lats -rt 01`
- Test all planets at all latitudes in the year 2003:
  `py PPchart.py -pv -lats -rt 03`
- Test first five planets at all latitudes in the years 2003 to 2004:
  `py PPchart.py -pv -lats -rt 03-04`

Required sample folder structure:

- Top level folder
    - _2000
    - _2001
    - _2002
    - _2003
    - _2004
    - code_execution folder
        - folder with source files in which to execute PPchart code
            - _rt   (folder with regression test results)

Differences, if any, are collected in a brief text file similar to "differences 2003.txt".
The following abbreviated example shows a perfect result with no discrepancies:

```
------ Process Mercury: 2003 at latitude 60.0°S ------
------ Process Mercury: 2003 at latitude 58.0°S ------
------ Process Mercury: 2003 at latitude 55.0°S ------
------ Process Mercury: 2003 at latitude 50.0°S ------
------ Process Mercury: 2003 at latitude 45.0°S ------
------ Process Mercury: 2003 at latitude 40.0°S ------
------ Process Mercury: 2003 at latitude 30.0°S ------
------ Process Mercury: 2003 at latitude 15.0°S ------
------ Process Mercury: 2003 at latitude 0.0°N ------
------ Process Mercury: 2003 at latitude 15.0°N ------
. . . . etc. . . . .
------ Process Mercury: 2003 at latitude 70.0°N ------
------ Process Mercury: 2003 at latitude 71.0°N ------
------ Process Mercury: 2003 at latitude 72.0°N ------

------ Process Venus: 2003 at latitude 60.0°S ------
------ Process Venus: 2003 at latitude 58.0°S ------
------ Process Venus: 2003 at latitude 55.0°S ------
. . . . etc. . . . .
```